

2024



Reducing Processing Time by 11x and Cost by 90% Using AWS Step Functions at Vertex Pharmaceuticals

Learn how Vertex Pharmaceuticals boosted image segmentation speed by 11 times using AWS Step Functions Distributed Map.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

11x

Accelerated image segmentation



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Overview

[Vertex Pharmaceuticals](#) (Vertex) is a global biotechnology company that invests in scientific innovation to create transformative medicines for people with serious diseases. To aid scientific research, scientists use image segmentation, a technique that parses pictures into discrete groups of pixels and uses machine learning to identify areas of interest. A single experiment could result in thousands of images that highly



to carefully examine as they identify and measure the results. To save hours of scientists' valuable time, Vertex uses Amazon Web Services (AWS) serverless technologies to automate and accelerate image segmentation.

Vertex's serverless architecture analyzes images at scale, implementing [AWS Step Functions](#), a visual workflow service that lets developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and ML pipelines. Using [AWS Step Functions Distributed Map](#) for large-scale parallel data processing, Vertex has accelerated experiment image analysis by 11 times while reducing costs by 90 percent, helping scientists iterate on their research more quickly in their search for transformative therapeutics.



Opportunity | Using AWS Step Functions to Modernize Image Segmentation for Vertex

Vertex scientists use ML to quicken the time-consuming, meticulous processes that are involved in analyzing microscope images of biologic samples from experiments. After designing an experiment, they manually trained the base ML model about what they hoped to identify from the images, labeling each area of interest by hand. After progressively fine-tuning the ML model on several images, the scientists ran the model across the rest of the images as a batch.

In the legacy system architecture, all scientists shared a single pool of compute resources. They sometimes had to wait up to 4 hours for results as their images queued for processing. “The image segmentation workload is incredibly spiky, and we continuously had to scale our compute fleet to match peak demand across scientific groups,” says Roberto Iturralde, senior director of software engineering at Vertex. “Under heavy load, performance degraded; under low load, we were overprovisioned.” To accommodate unpredictable demand, Vertex had to keep its servers running 24-7, accumulating cost.

Working alongside AWS, Vertex redesigned its system in April 2023 using AWS Step Functions Distributed Map and [AWS Lambda](#), a serverless, event-driven compute service that lets engineers run code for virtually any application or backend service without provisioning or managing servers. Vertex uses infrastr



model and deploy its system through its continuous integration and deployment pipelines so that Vertex engineers can safely and quickly deploy changes to the image segmentation application.

As an early adopter of AWS Step Functions Distributed Map, Vertex can run up to 10,000 parallel workflows. It can also iterate from millions of items in various formats that are stored in [Amazon Simple Storage Service \(Amazon S3\)](#), an object storage service offering scalability, data availability, security, and performance. “Since the new system went into production, we have hardly had to think about it,” says Iturralde. “Because it’s serverless, it just runs.”

Solution | Increasing Processing Speed by 11 Times and Cutting Costs by 90 Percent for Image Segmentation Using AWS Serverless Technologies

Scientists interact with a web application running on [AWS Fargate](#), a serverless, pay-as-you-go compute engine that lets engineers focus on building applications without managing servers. Using the web application, scientists fine-tune the parameters of the ML model that segments the image and test the model on individual images. Using AWS serverless technologies, the subsequent batch of images now process automatically. The automated process begins with a call to Amazon API Gateway, a fully managed service that makes it simple for developers to create, publish, maintain, monitor, and secure APIs at nearly any scale. The AWS Step Functions workflow uses the input from [Amazon API Gateway](#) to determine which images to process and which parameters to pass back to the ML model. Using AWS Step Functions Distributed Map, the system processes all the images in parallel. Finally, the system automatically writes the output to Amazon S3, recording the metadata in a database for review.

Using AWS Step Functions, Vertex accelerated performance by 11 times because workflows now take minutes instead of hours. “Initially, some of our scientists questioned if the system was working, because it was going so fast compared with the old one,” says Iturralde.

Using the scalability of AWS Step Functions Distributed Map and AWS Lambda, dozens of scientists can work simultaneously and the system runs across multiple [AWS Availability Zones](#). “Using AWS Step Functions, we greatly improved the robustness and reliability of our image segmentation solution,” says Karthik Ghantasala, director of cloud architecture and strategy at Vertex. “Our scientists analyze their experiments quickly and cost effectively.” Because the serverless system runs only when it’s in use, Vertex has cut compute costs by 90 percent.



Vertex engineers have improved observability as thousands of images are processed across multiple experiments. “Using AWS Step Functions, we can pinpoint where exactly an error has caused a workflow to pause,” says Ghantasala. “We have also reduced operational overhead.” AWS Step Functions features native automatic retry and error-handling functionality, restarting from the failed step rather than from the beginning, which saves additional time and cost. To handle the observability, the team incorporates AWS monitoring tools such as [Amazon CloudWatch](#), a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health.

Outcome | Using AWS Step Functions Distributed Map for Other Workloads

Vertex has identified other areas where it can apply its learnings from AWS Step Functions Distributed Map. For example, during the analysis of a large library of scientific compounds, one unit of work spawns parallel units that need to be aggregated.

“Using AWS Step Functions, we can tackle our scientists’ issues,” says Iturralde. “We’re here to accelerate research and help bring safe, effective medicines to patients quickly.”

About Vertex Pharmaceuticals

Vertex invests in scientific innovation to create transformative medicines for people with serious diseases. It has approved therapies for cystic fibrosis, sickle cell disease, and transfusion-dependent beta thalassemia and has a robust research and development pipeline.

AWS Services Used

AWS Step Functions

AWS Step Functions is a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages compute resources, making it the fastest way to turn an idea into a modern, production, serverless application.



[Learn more »](#)

Amazon Simple Storage Service

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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Amazon API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

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